Adaptive Hybrid Control Error for Vehicle Platoons via Reinforcement Learning

Mid-Report

Authors:
Mustafa Kamal
Rafael Maestre López
Charelle Khoury

Course:
Project S8 - P22

Contents

1 Introduction **1**
1.1 A Short Description . . . 1
1.2 Motivation . . . 1
1.3 Our contributions . . . 1

2 Traffic Environment **1**
2.1 Driver Models . . . 1
2.1.1 Intelligent Driver Model (IDM) . . . 1
2.1.2 Stochastic Intelligent Driver Model (SIDM) . . . 2
2.1.3 Connected Autonomous Vehicle (CAV) Model . . . 2
2.2 Mixed Traffic Model . . . 3
2.3 Macroscopic Information . . . 4

3 Reinforcement Learning **4**
3.1 Fundamentals . . . 5
3.2 Reward design . . . 5
3.3 PPO . . . 6
3.4 SAC . . . 6
3.5 Role of RL in the proposed framework . . . 7

4 System Architecture **7**
4.1 Overall structure . . . 7
4.2 Simulation layer . . . 8
4.3 Mesoscopic perception layer . . . 8
4.4 Mesoscopic adaptation layer . . . 8
4.5 Control layer . . . 9

5 Experimental Results and Perspectives **9**
5.1 Protocol and admissibility fix . . . 9
5.2 Results . . . 10
5.3 Interpretation . . . 10
5.4 Future work . . . 11

1 Introduction

1.1 A Short Description

The modernization of intelligent transportation systems (ITS) has necessitated a paradigm shift from individual vehicle autonomy to cooperative automation. Vehicular platooning offers a promising solution to traffic congestion, fuel inefficiency, and road safety.

Thus, our work motivates an "Adaptive Hybrid Control strategy" using RL to adjust the weight between micro and macro. This transition from the standard Adaptive Cruise Control (ACC) to Cooperative Adaptive Cruise Control (CACC) introduces challenges in string stability, the ability to attenuate disturbances, especially under communication delays.

1.2 Motivation

A fundamental trade-off exists in platoon control architectures:

- **Local Control:** Relies on local sensors (radar/LiDAR). It is robust to network failures and ensures collision avoidance but is prone to string instability in long platoons.
- **Global Control:** Relies on V2V data from the leader. It provides excellent string stability but is highly sensitive to communication delays and packet loss.

1.3 Our contributions

- Modeling mixed traffic with human-driven(HDVs) and connected autonomous vehicles (CAVs), improving capacity and stability, in addition to extending traffic analysis from open platoons to closed ring-road systems.
- Proposing a mesoscopic RL framework combining micro- and macro-level traffic states with incorporated resistance forces on our input to create the nearest real life simulation.

2 Traffic Environment

2.1 Driver Models

2.1.1 Intelligent Driver Model (IDM)

The Intelligent Driver Model (IDM) is a deterministic microscopic car-following model describing the longitudinal dynamics of vehicle i as a function of its own velocity, the spacing to its predecessor, and their relative velocity [1]. The acceleration of vehicle i is defined as

$$\dot{v}_i = a_{max}\left[\underbrace{1-\left(\frac{v_i}{v_d}\right)^{\delta}}_{\text{free-road acceleration}} - \underbrace{\left(\frac{s^*(v_i,\Delta v_i)}{s_i}\right)^2}_{\text{interaction (braking) term}}\right]. \tag{1}$$

v_i: velocity of vehicle i,
v_d: the desired free-flow velocity
a_{max}: the maximum acceleration
δ: an acceleration exponent controlling how smoothly the vehicle approaches the desired velocity in free traffic.

When $v_i \ll v_0$, the free-road acceleration $\simeq 1$, so the vehicle accelerates approximately with magnitude a_{max}. As v_i approaches v_0, the term smoothly converges to zero, ensuring bounded acceleration and asymptotic convergence to the desired free-flow equilibrium [1].

The interaction term dominates when the actual spacing s_i is less than the desired spacing s^*, thereby inducing decceleration. The quadratic structure ensures strong braking for small gaps and guarantees collision-free behavior under realistic parameter choices [1].

The spacing is defined as

$$s_i = x_{i-1} - x_i - L, \tag{2}$$

where x_i and x_{i-1} denote the longitudinal positions of vehicle i and its leader, respectively, and L is the vehicle length. The relative velocity is defined as $\Delta v_i = v_i - v_{i-1}$.

The desired dynamic spacing is given by

$$s^*(v_i, \Delta v_i) = \underbrace{s_0}_{\text{jam distance}} + \underbrace{v_i T_H}_{\text{time-headway term}} + \underbrace{\frac{v_i \Delta v_i}{2\sqrt{a_{max} b}}}_{\text{anticipation term}} . \tag{3}$$

The constant term s_0 represents the minimum spacing in standing traffic. The linear term $v_i T_H$ reflects the constant time-headway policy characteristic of human driving behavior. The third term, proportional to $v_i \Delta v_i$, introduces anticipative braking by accounting for relative velocity differences and the maximum acceleration and desired deceleration denoted respectively by a_{max} and b [1].

2.1.2 Stochastic Intelligent Driver Model (SIDM)

To account for perception errors, behavioral variability, and unmodeled disturbances, the IDM can be extended by introducing a stochastic acceleration component [2]. The stochastic IDM (SIDM) is formulated as

$$\dot{v}_i = \underbrace{f(s_i, v_i, v_{i-1})}_{\text{deterministic IDM}} + \underbrace{\xi_i(t)}_{\text{stochastic acceleration noise}} , \tag{4}$$

where $f(\cdot)$ denotes the deterministic IDM acceleration function and $\xi_i(t)$ is a stochastic process representing white acceleration noise.

The stochastic term satisfies

$$\langle \xi_i(t) \rangle = 0, \tag{5}$$

$$\langle \xi_i(t) \xi_j(t') \rangle = Q\, \delta_{ij}\, \delta(t - t'), \tag{6}$$

where Q denotes the noise intensity, δ_{ij} is the Kronecker delta ensuring independence between different vehicles, and $\delta(t-t')$ is the Dirac delta distribution representing temporal white noise [2].

Close to the string stability threshold, noise can generate spatiotemporal correlations and traffic oscillations resembling stop-and-go waves. Consequently, the SIDM provides a unified framework for analyzing deterministic instabilities and noise-induced traffic dynamics [2].

2.1.3 Connected Autonomous Vehicle (CAV) Model

For connected autonomous vehicles, we adopt the point-mass longitudinal dynamics described in [3]. The motion of vehicle i is governed by

$$\dot{x}_i = v_i, \qquad \dot{v}_i = u_i, \tag{7}$$

In a car-following configuration, the spacing and relative velocity are defined as

$$s_i = x_{i-1} - x_i - L, \qquad \Delta v_i = v_i - v_{i-1}. \tag{8}$$

CAVs implement an Adaptive Cruise Control (ACC) strategy based on a Constant Time Headway (CTH) policy. The desired spacing is given by

$$s_i^{\text{ref}} = s_0 + T_C v_i, \tag{9}$$

where s_0 denotes the standstill spacing and $T_C > 0$ is the selected time headway.

Defining the spacing error

$$e_{p,i} = s_i - s_i^{\text{ref}}, \tag{10}$$

its dynamics follow from

$$\dot{s}_i = -\Delta v_i, \qquad \dot{s}_i^{\text{ref}} = T_C \dot{v}_i = T_C u_i, \tag{11}$$

which yields

$$\dot{e}_{p,i} = -\Delta v_i - T_C u_i. \tag{12}$$

A linear ACC controller that regulates both spacing and velocity errors can therefore be written as

$$u_i = k_s(s_i - s_i^{\text{ref}}) - k_v \Delta v_i, \tag{13}$$

where $k_s > 0$ and $k_v > 0$ denote the spacing and velocity feedback gains, respectively. This control law constitutes a linear realization of the CTH policy and is consistent with the mesoscopic ACC formulation proposed in [3].

An important property of the CTH-based ACC strategy is its capability to guarantee weak string stability using only local measurements [4]. In the frequency domain, the disturbance propagation transfer function satisfies

$$\|H(j\omega)\|_\infty \leq 1, \tag{14}$$

ensuring that spacing disturbances are not amplified upstream along a vehicle string. Moreover, robustness with respect to sensing and actuation delays imposes a constraint on the admissible implementation lag. In particular, the total delay must satisfy

$$T_C \leq \frac{h_w}{2}, \tag{15}$$

indicating that allowable delay is directly limited by the chosen time headway [4].

2.2 Mixed Traffic Model

We consider a mixed traffic system composed of N vehicles circulating on a single-lane ring road of length L_{ring}. The vehicle indices satisfy periodic boundary conditions such that vehicle 1 follows vehicle N.

Let $x_i(t)$ and $v_i(t)$ denote the position and velocity of vehicle $i \in \{1, \ldots, N\}$. The ring-road constraint is given by

$$x_{i-1}(t) > x_i(t), \qquad x_0(t) := x_N(t) + L_{\text{ring}}. \tag{16}$$

The spacing is defined as

$$s_i = x_{i-1} - x_i - L, \tag{17}$$

where periodicity applies for $i = 1$.

Let $\mathcal{H}$ denote the set of HDVs and $\mathcal{C}$ the set of CAVs, with penetration rate

$$P = \frac{|\mathcal{C}|}{N}. \tag{18}$$

Human-driven vehicles are modeled according to the stochastic Intelligent Driver Model introduced in Subsection 2.1.2.

HDVs do not utilize acceleration information of the preceding vehicle, reflecting the absence of communication capabilities. Furthermore, HDVs are characterized by a larger effective reaction time T_H, representing the delay associated with human perception and actuation. Consequently, their response to traffic perturbations is slower and more susceptible to amplification of stochastic fluctuations.

Connected autonomous vehicles are modeled according to the ACC framework introduced previously in 2.1.3. In contrast to HDVs, CAVs are assumed to exhibit higher sensitivity coefficients and a shorter desired time headway T_C, reflecting their faster sensing [5]. The condition $T_C < T_H$ therefore captures the higher responsiveness of CAVs compared to HDVs.

When a CAV follows an HDV, communication-based acceleration information is unavailable. In this degraded configuration, the feedforward gain is set to $k_a = 0$, and the effective time headway is increased to $T_C^{\text{deg}} > T_C$, representing reduced cooperative capability. This modeling choice captures the performance degradation of CAVs in mixed traffic reported in [5] and [6].

The closed-loop mixed system is therefore

$$\dot{v}_i = \begin{cases} f_{\text{IDM}}(s_i, v_i, v_{i-1}) + \xi_i(t), & i \in \mathcal{H}, \\ k_s(s_i - s_i^{\text{ref}}) - k_v \Delta v_i + k_a a_{i-1}, & i \in \mathcal{C}. \end{cases} \tag{19}$$

2.3 Macroscopic Information

Up to this point, only microscopic vehicle interactions have been considered. To improve the performance of the proposed framework, macroscopic traffic information are introduced according to [3]. In particular, we define the mean velocity on the ring as

$$\bar{v}(t) = \frac{1}{N} \sum_{i=1}^{N} v_i(t), \tag{20}$$

and the velocity variance

$$\sigma_v^2(t) = \frac{1}{N} \sum_{i=1}^{N} \left(v_i(t) - \bar{v}(t)\right)^2. \tag{21}$$

The mean velocity represents the macroscopic traffic state, while the variance quantifies the level of speed dispersion and therefore the intensity of oscillations. As discussed in [3], the variance provides a compact indicator of transient traffic behavior and string instability.

3 Reinforcement Learning

In this project, reinforcement learning (RL) is used to improve the behavior of connected and automated vehicles (CAVs) in mixed traffic. The main idea is not to replace the baseline

controller, but to add an adaptive layer on top of it. This learned layer adjusts the time headway policy according to the current traffic situation. In this way, the controller keeps its physical interpretation while gaining the ability to react to complex traffic patterns that are difficult to model explicitly.

3.1 Fundamentals

The RL problem is formulated as a Markov Decision Process (MDP),

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r),$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, P represents the traffic dynamics, and r is the reward.

At each time step, the agent observes the traffic state, selects an action, applies it to the controller, and receives a reward. In the current implementation, the state of each CAV i is defined as

$$x_i = (\mu_{v,i}, \sigma^2_{v,i}, \delta^{\text{macro}}_{v,i}, v_i, s_i, \Delta v_i, a_{\ell(i)}, \alpha_i^{prev}),$$

where $\mu_{v,i}$ and $\sigma^2_{v,i}$ are the upstream mean velocity and velocity variance, $\delta^{\text{macro}}_{v,i} = v_i - \mu_{v,i}$ is the speed mismatch with the local upstream flow, v_i is the ego speed, s_i is the spacing to the leader, Δv_i is the relative speed, $a_{\ell(i)}$ is the leader acceleration, and α_i^{prev} is the previously applied headway scaling factor.

This state combines *microscopic* information, such as spacing and relative velocity, with *mesoscopic* information, such as the average speed and speed variance of upstream traffic. This is important because local measurements alone may not be enough to detect the development of traffic waves.

The action is a residual correction of the headway scaling:

$$\mathcal{A} = \{\Delta\alpha_i \in [-\Delta\alpha_{\max}, +\Delta\alpha_{\max}]\}.$$

The final headway scaling factor is then given by

$$\alpha_i(t) = \text{clip}\left(\alpha_i^{rule}(t) + \Delta\alpha_i(t)\right),$$

where $\alpha_i^{rule}(t)$ is the baseline headway scaling provided by the rule-based mesoscopic module. Therefore, RL only refines the baseline decision instead of replacing it. The learned policy can be written as

$$\pi : \mathcal{S} \to \mathcal{A}, \qquad \Delta\alpha_i(t) = \pi(x_i(t)).$$

3.2 Reward design

The reward function is designed to encourage safe, smooth, and efficient traffic behavior while improving string stability. The total reward is

$$r_t = r_{\text{sf}}[t] + r_{\text{sp}}[t] + r_{\text{ef}}[t] + r_{\text{cf}}[t] + r_{\text{ss}}[t] + r_{\sigma^2}[t].$$

The desired spacing is defined by

$$s_i^{des}[t] = d_0 + h_i'[t]\, v_i[t],$$

and the spacing error is

$$e_i[t] = s_i[t] - s_i^{des}[t].$$

The different reward terms have a clear physical meaning:

- r_{sf}: safety term, penalizing unsafe spacing and too small time headway;
- r_{sp}: spacing regulation term, encouraging tracking of the desired spacing;
- r_{ef}: efficiency term, encouraging the vehicle speed to remain close to the upstream mean speed;
- r_{cf}: comfort term, penalizing high jerk;
- r_{ss}: string stability term, penalizing amplification of spacing errors along the platoon;
- r_{σ^2}: traffic smoothness term, penalizing large velocity variance in the traffic stream.

This reward formulation is useful because it combines several objectives into a single learning signal. It allows the RL agent to balance local vehicle behavior and global traffic quality.

3.3 PPO

Proximal Policy Optimization (PPO) is a policy-gradient RL algorithm that updates the policy in a stable and conservative way. Its main advantage is that it is simple to implement and usually robust during training. PPO avoids large policy updates by using a clipped objective:

$$L^{\mathrm{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right],$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t|s_t)},$$

and $\hat{A}_t$ is the estimated advantage.

For our problem, PPO is attractive because the action is continuous and the controller must remain stable during learning. It is a good candidate for a first implementation because of its simplicity and reliability.

3.4 SAC

Soft Actor-Critic (SAC) is another RL algorithm for continuous control. Unlike PPO, SAC is an off-policy method and includes an entropy term in its objective:

$$J(\pi) = \sum_t \mathbb{E}\left[r(s_t, a_t) + \beta \mathcal{H}(\pi(\cdot|s_t)) \right],$$

where $\mathcal{H}(\pi(\cdot|s_t))$ is the policy entropy and β controls the exploration level.

The main interest of SAC is its good sample efficiency and its strong performance in continuous-action problems. Since our action is a continuous residual correction $\Delta\alpha_i$, SAC is also well suited to this framework. However, compared with PPO, SAC is usually more complex to tune and explain. We also chose to consider SAC because the mixed-traffic environment is stochastic due to HDV behavior. SAC learns from a replay buffer containing past interactions, which allows it to reuse previous experience instead of relying only on newly collected data. This can make learning more sample-efficient and potentially more robust to traffic variability. In contrast, PPO uses mainly the data generated by the current policy at each training iteration. For this reason, PPO is often easier to implement, while SAC is an interesting alternative for noisy and stochastic environments.

3.5 Role of RL in the proposed framework

The role of RL in our approach is to make the controller adaptive while preserving the original control structure. The baseline controller still ensures a physically meaningful longitudinal behavior, while RL improves the headway adaptation using both microscopic and mesoscopic traffic information.

This integration has three main advantages. First, it keeps the controller interpretable for a control-theory audience. Second, it allows the CAV to react not only to its immediate leader, but also to collective upstream traffic conditions. Third, it avoids solving a complex optimization problem online at each time step. For these reasons, RL is a suitable tool for improving stability and smoothness in mixed traffic.

4 System Architecture

The proposed system is organized as a layered control architecture for mixed traffic on a ring road. Its main objective is to combine microscopic control with mesoscopic traffic awareness. The complete framework can be divided into four main parts: the simulation layer, the mesoscopic perception layer, the mesoscopic adaptation layer, and the control layer. Each part has a specific role, and together they allow the CAVs to react not only to their immediate leader but also to the collective traffic conditions upstream.

4.1 Overall structure

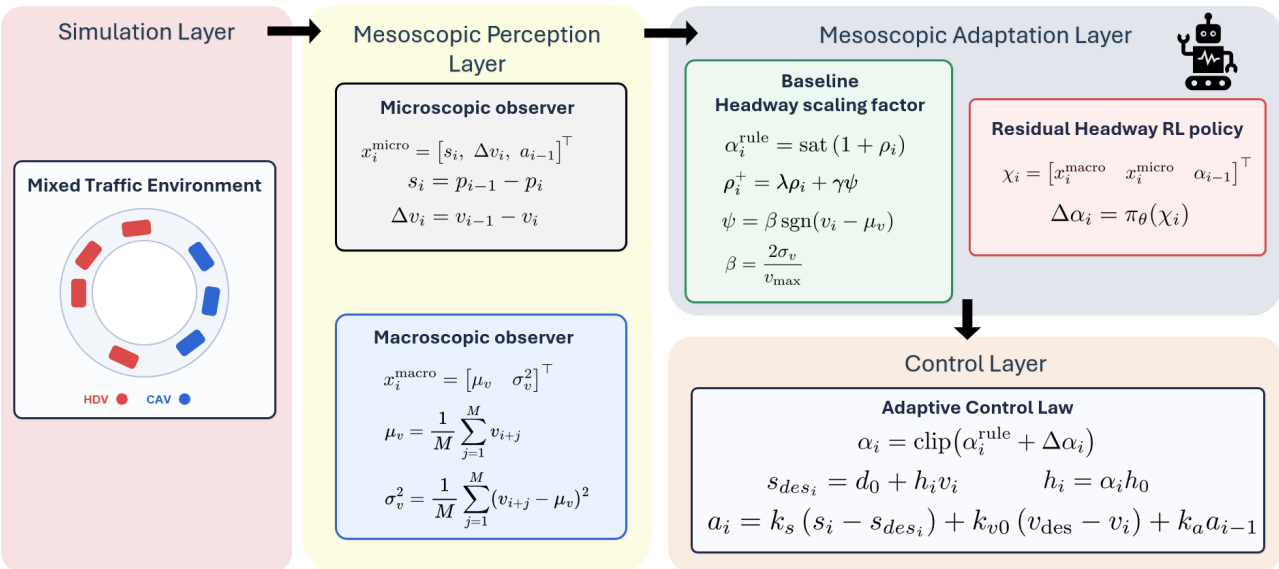

Figure 1: Overview of the proposed system architecture.

The simulation layer represents the mixed-traffic environment, where human-driven vehicles (HDVs) and connected and automated vehicles (CAVs) evolve on the same road. At each time step, the CAV receives observations from the traffic environment, processes them through two perception channels, computes an adaptive parameter, and finally applies the resulting control law. In this way, the architecture separates *observation*, *adaptation*, and *actuation*, which makes the framework easier to interpret from a control perspective.

4.2 Simulation layer

The simulation layer contains the traffic environment and generates the variables needed by the control framework. The road is modeled as a ring road, which avoids boundary effects and makes the appearance of traffic oscillations easier to study. HDVs and CAVs coexist in the same environment, but only the CAVs use the adaptive control architecture.

This layer provides the real-time traffic variables used by the upper layers, such as positions, velocities, spacing, and acceleration. It is therefore the source of both local and collective traffic information.

4.3 Mesoscopic perception layer

The perception layer extracts the information used by the controller. It contains two complementary observers.

Microscopic observer. The microscopic observer describes the local car-following situation of vehicle i. It uses the spacing, relative velocity, and ego speed:

$$x_i^{\text{micro}} = \begin{bmatrix} s_i & \Delta v_i & a_{i-1} \end{bmatrix}^\top,$$

with

$$s_i = p_{i-1} - p_i, \qquad \Delta v_i = v_{i-1} - v_i.$$

These variables describe the immediate interaction with the preceding vehicle and are the standard quantities used by longitudinal controllers.

Macroscopic observer. The second observer captures collective traffic behavior upstream of the CAV. Instead of looking only at the leader, it summarizes the traffic state over a perception horizon. The two main quantities are the upstream mean velocity and the upstream velocity variance:

$$x_i^{\text{macro}} = \begin{bmatrix} \mu_v & \sigma_v^2 \end{bmatrix}^\top,$$

with

$$\mu_v = \frac{1}{M}\sum_{j=1}^{M} v_{i+j}, \qquad \sigma_v^2 = \frac{1}{M}\sum_{j=1}^{M}(v_{i+j} - \mu_v)^2.$$

The mean velocity gives an estimate of the average upstream traffic speed, while the variance indicates how irregular or unstable the traffic stream is.

4.4 Mesoscopic adaptation layer

The adaptation layer transforms the observed traffic information into a headway modification for the CAV controller. It contains two parts: a baseline adaptive headway mechanism and a residual reinforcement learning policy.

Baseline adaptive headway. The baseline mechanism computes a rule-based headway scaling factor from the macroscopic traffic information. The update can be written as

$$\alpha_i^{\text{rule}} = \text{sat}(1 + \rho_i),$$

with the internal state

$$\rho_i^+ = \lambda \rho_i + \gamma \psi,$$

and the perception signal

$$\psi = \beta\,\mathrm{sgn}(v_i - \mu_v), \qquad \beta = \frac{2\sigma_v}{v_{\max}}.$$

This mechanism has a simple interpretation. When upstream traffic becomes more variable, the controller increases the effective headway in order to react more cautiously. When traffic is smoother, the headway can be reduced. The filtering through ρ_i also avoids abrupt parameter changes.

Residual RL policy. On top of the baseline adaptation, a residual policy computes a correction term:

$$\chi_i = \begin{bmatrix}\mu_v & \sigma_v^2 & (v_i - \mu_v) & v_i & s_i & \Delta v_i & a_{i-1} & \alpha_i^{prev}\end{bmatrix}^\top, \qquad \Delta\alpha_i = \pi_\theta(\chi_i).$$

The role of this block is to refine the rule-based decision without changing the overall structure of the controller. In other words, the adaptive law remains physically interpretable, while the learning block adds flexibility when the traffic situation becomes more complex. The resulting policy input therefore combines mesoscopic information, local car-following variables, and the previously applied headway state in a single 8-dimensional vector.

4.5 Control layer

The control layer converts the adapted headway into a longitudinal acceleration command for the CAV. The final scaling factor is

$$\alpha_i = \mathrm{clip}\big(\alpha_i^{\mathrm{rule}} + \Delta\alpha_i\big)\,,$$

and the corresponding time headway becomes

$$T_i = \alpha_i T_0.$$

This adapted headway is then used inside the microscopic CACC/CTH controller:

$$a_i = k_s\,(s_i - d_0 - T_i v_i) + k_v \Delta v_i + k_a a_{i-1} + k_{v0}(v_{\mathrm{des}} - v_i).$$

This equation shows clearly how the architecture is organized. The lower-level controller remains a classical feedback law based on spacing, relative speed, leader acceleration, and desired speed regulation. The upper layers do not replace this controller; they only adapt the time-headway parameter T_i used inside it.

5 Experimental Results and Perspectives

5.1 Protocol and admissibility fix

The experiment targets the 50 % human-driver regime on a ring road with N=22 vehicles (L=300 m). Three methods are compared: the rule-based mesoscopic adaptation alone (**No RL**), and two residual RL policies (**PPO**, **SAC**) trained on top of it. Each episode starts from random initial conditions; a velocity perturbation ($\Delta v = -3$ m/s) is applied at t=12 s after a 10 s warm-up. The primary metric is the *string-stability amplification ratio*: the maximum downstream-to-source speed-error ratio, where a value below 1 indicates disturbance attenuation.

A critical protocol bug was found and fixed before this campaign. The random initialization computed minimum inter-vehicle spacing without accounting for vehicle length (ℓ=5 m), producing net bumper-to-bumper gaps as low as 0 m. Combined with a warm-up that also capped

braking, every prior run suffered collision-clamp events *before* the perturbation, invalidating string-stability. The fix added vehicle length to the spacing bound, clamped initial closing rates to be kinematically safe, and allowed unrestricted braking during warm-up. After the fix, 50/50 random seeds produced zero pre-perturbation clamps.

5.2 Results

Both RL methods early-stopped quickly (PPO at episode 15, SAC at 21), with best checkpoints at episodes 4 and 6. This reflects the limited residual authority ($|\Delta\alpha| \leq 0.1$): the policy reaches its useful range almost immediately and subsequent updates add noise. Zero collision clamps were observed throughout training, confirming the initialization fix.

Table 1 presents the evaluation. All three runs are fully admissible: zero collisions, zero clamps, perturbation correctly applied, and valid string-stability measurement.

Table 1: Evaluation at 50 % human ratio. All runs admissible. Bold = best per column.

Method	Amp. ratio (–)	Var_{100} ($\mathrm{m}^2/\mathrm{s}^2$)	Mean spd_{100} (m/s)	RMS acc (m/s^2)	Min gap (m)	Clamps
No RL	0.873	0.517	**4.408**	**0.399**	2.875	**0**
PPO	0.881	0.687	4.183	0.403	2.929	**0**
SAC	**0.873**	**0.482**	4.325	0.425	**2.957**	**0**

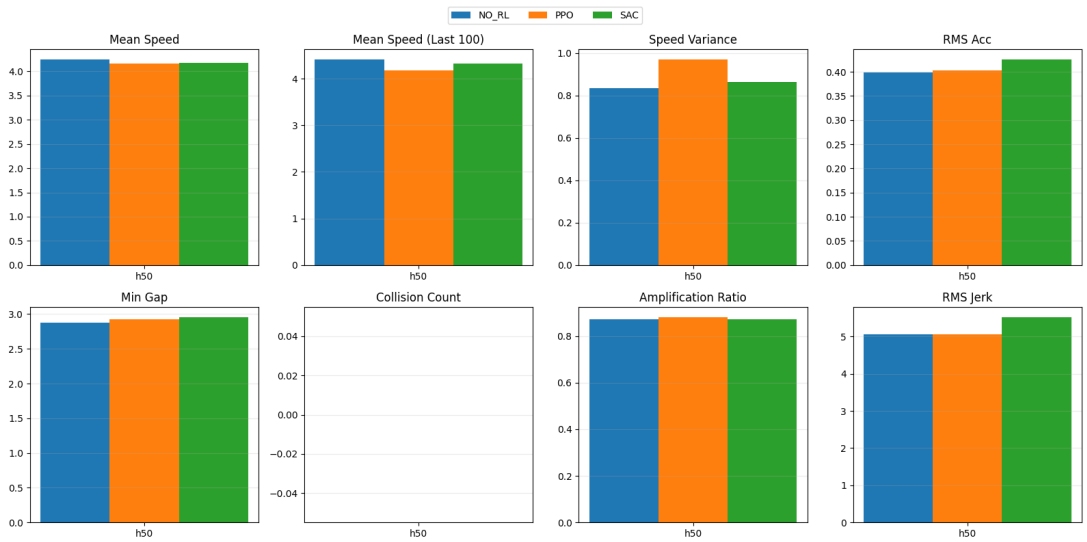

Figure 2: Bar comparison of evaluation metrics. SAC achieves the lowest late-regime speed variance and the largest minimum gap.

5.3 Interpretation

The mesoscopic baseline is already strong. With an amplification ratio of 0.873, the rule-based adaptation alone produces string-stable flow, validating the control architecture independently of learning.

The RL residual has limited marginal impact. SAC matches the baseline on the primary metric while reducing late-regime speed variance by 7 % and improving the minimum gap. PPO performs slightly worse than the baseline, consistent with lower sample efficiency in a low-authority regime. With only ±10 % headway modulation, neither algorithm deviates far from the rule-based policy.

SAC outperforms PPO. Across all secondary metrics (speed variance, minimum gap), SAC is consistently better. Its entropy-regularised objective appears better suited to continuous-action tasks with limited exploration budgets.

The protocol is now valid. Unlike all prior campaigns, this run satisfies the admissibility gate, making the comparison scientifically meaningful for the first time.

5.4 Future work

The most immediate next step is **multi-seed evaluation** (10–20 seeds) to distinguish genuine method differences from seed artefacts. Beyond that, **increasing the RL authority** ($|\Delta\alpha|$ from 0.1 to 0.2–0.3) would give the policy more room to differentiate from the baseline. The campaign should also be extended to **25 % and 75 % penetration rates**, and training should move to the **GPU-vectorized backend** to enable longer runs with larger batch sizes.

References

[1] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," 2000.

[2] M. Treiber and A. Kesting, "The intelligent driver model with stochasticity – new insights into traffic flow oscillations," *Transportation Research Part B: Methodological*, vol. 117, pp. 613–623, 2018.

[3] M. Mirabilio, A. Iovine, E. de Santis, M. D. Di Benedetto, and G. Pola, "A mesoscopic human-inspired adaptive cruise control for eco-driving," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 9, pp. 9571–9583, 2023.

[4] D. Swaroop and K. R. Rajagopal, "A review of constant time headway policy for automatic vehicle following," in *ITSC 2001. 2001 IEEE Intelligent Transportation Systems. Proceedings (Cat. No.01TH8585)*, pp. 65–69, IEEE, 2001.

[5] Y. Gong and W.-X. Zhu, "Modeling and robust h ∞ control synthesis of the cav-hdv heterogeneous traffic system with different car-following modes," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 10, pp. 12980–12998, 2024.

[6] B.-l. Cen, Y. Xue, K. Zhang, L. Jia, and H.-d. He, "Study on traffic flows with connected vehicles and human-driven vehicles," *Applied Mathematics and Computation*, vol. 490, p. 129182, 2025.